Alok Raj

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Education

Vellore Institute of Technology Sep 2023 – May 2027
B.Tech in Computer Science and Engineering (Cloud Computing and Automation) *CGPA: 8.93/10*

Technical Skills

Languages: Java, Python, SQL
Cloud: AWS (Lambda, EC2, API Gateway, S3, DynamoDB, EventBridge, CloudWatch, IAM, Bedrock, Rekognition)
Backend: REST APIs, FastAPI, Microservices
GenAI: LangChain, RAG, Vector Databases (Pinecone, OpenSearch), Embeddings
Databases: MySQL, MongoDB, DynamoDB
Tools: Git, GitHub Actions, Terraform, Docker, n8n, Postman

Experience

Cloud Engineering Intern Aug 2025 – Jan 2026
IPAGE Group *Remote*

- Built and deployed 20+ AWS Lambda functions behind API Gateway to automate an 8-step company registration and AI content-generation workflow, designed to support 1,000 concurrent users and 500–1,000 daily registrations.
- Reduced AI content-generation cost by 88% by routing each request to GPT-3.5 or GPT-4 based on section type and input richness, measured by per-generation token consumption.
- Designed a three-stage approval pipeline (draft, under review, approved) with DynamoDB state tracking and Nodemailer notifications at each transition, replacing manual follow-up.
- Implemented wallet token deduction for AI generation billing: validated balance before generation, enforced idempotency to prevent double deduction, and kept token state consistent across two data stores.

Projects

MarketMind | RAG-powered stock intelligence platform | GitHub Feb 2026 – Mar 2026
Tech Stack: Python, FastAPI, Groq (LLaMA), React, Pinecone, BAAI/bge-base-en-v1.5 embeddings

- Built a RAG-powered stock intelligence platform with a 3-member team, ingesting data from five live financial sources for semantic retrieval and real-time market signal analysis.
- Designed and built the Finfluencer Fact-Checker: extracts financial claims from YouTube transcripts chunk by chunk, verifies each against live market data, and returns a verdict (Verified, Misleading, False, Unverifiable) with a video-level risk score.
- Wireframed the platform's UI and integrated it with APIs across all features, delivering a seamless end-to-end user experience.

Sahayak AI | Multilingual Autonomous Teaching Assistant | GitHub Sep 2025 – Jan 2026
Tech Stack: AWS Lambda, Amazon Bedrock, OpenSearch Serverless, S3, EventBridge, DynamoDB

- Developed a classroom automation platform for Indian middle schools (Grades 6–8) using AWS Lambda functions and Amazon Bedrock (Nova, Titan), automating lesson scheduling and student doubt resolution.
- Built an NCERT knowledge base on S3 with OpenSearch Serverless vector search and a multilingual RAG pipeline using Amazon Translate and Bedrock for curriculum-grounded doubt resolution.
- Implemented an EventBridge-driven pipeline that enriches teacher-uploaded lessons with Bedrock-generated questions and YouTube recommendations, delivers them to students, and sends notifications via Nodemailer.

SafeX 3.0 | 3-factor biometric authentication system | GitHub Feb 2025 – Apr 2025
Tech Stack: AWS Lambda, API Gateway, Rekognition, S3, MongoDB, IAM, CloudWatch

- Designed a 3-factor authentication system (bcrypt password, AWS Rekognition face-match similarity threshold of 99%, OTP) across 8 REST endpoints on API Gateway, with face data in S3 and user metadata in MongoDB.
- Implemented IAM role-based access, presigned S3 URLs for the file vault, and CloudWatch logging across Lambda functions to support operational monitoring and event traceability.

Certifications and Achievements

- **AWS Certified Solutions Architect – Associate** – Jun 2026 (Score: 893/1000)
- **AWS Certified Cloud Practitioner** – Apr 2026 (Score: 1000/1000)
- **Top Performing Candidate** – ET AI Hackathon 2026, The Economic Times (May 2026)
- **Winner** – Nebula Nexus Hackathon, MIT Jaipur (Jul 2025)